



Technical Assessment — Customer Assistant for Kopano Microfinance

Junior Developer — Operations · Xavier Africa · In person · About 4 hours

Time breakdown

Time	Stage	What you do
0:00 – 0:10	Briefing	We walk you through this brief and the materials. Ask anything.
0:10 – 0:35	Discovery and planning	Read the materials, question the client, decide what you will build.
0:35 – 0:45	Checkpoint 1	Present your understanding of the problem and your plan.
0:45 – 1:05	Design	Work out how you will build it.
1:05 – 1:15	Checkpoint 2	Present your design.
1:15 – 3:15	Build and deploy	Two hours to implement and deploy.
3:15 – 3:25	Break	Final preparation.
3:25 – 3:55	Walkthrough	Demo your assistant and take our questions.

The client

Kopano Microfinance is a micro lender with four branches in Botswana, serving salaried workers, small business owners and families.

Their customer service team of four agents is overwhelmed. Most of what they handle is repetitive: customers asking for their balance, when their next payment is due, how to pay, and what happens if they pay late. Meanwhile, customers who have already paid are still being chased, because payments take days to be allocated to loans. Complaints are rising.

Kopano wants an AI assistant on their website that handles routine questions around the clock and hands customers over to a human agent whenever it should not be answering on its own.

Gaone Tsheko, Kopano's Head of Customer Operations, is your client for the day. One of the interviewers will play her. You can ask her questions at any time — she will answer what you ask, and nothing more.

What you receive

1. **kopano_data.xlsx** — an extract of Kopano's loan book, recent payments and collections contact history. **Start with the Read Me sheet.**
2. **knowledge_base/** — 11 of Kopano's policy and information documents, in a mix of PDF, Word and text files.
3. **An API key** for a large language model, with a spending limit.

Treat today's date as **Tuesday 6 October 2026**.

What to build

An AI chat assistant that Kopano's borrowers can use to get answers about their loans and Kopano's policies, backed by a human agent handover.

Core requirements — everyone builds these

1. **A chat interface** a customer could use. A web page is preferred; any interface you can demo is acceptable.
2. **Policy answers from the knowledge base.** The assistant answers questions about Kopano's products, policies and procedures using retrieval over the provided documents, and shows which document each answer came from.
3. **Identity verification.** Before sharing any account information, the assistant verifies the customer using their registered mobile number and the last four digits of their ID number.
4. **Account answers from the data.** For a verified customer, the assistant can answer at least: *when is my next payment due and how much is it*, and *what is my outstanding balance*. Figures must come from the data, never be generated by the model.
5. **Human handover.** The assistant hands the conversation to a human agent when it should, following Kopano's own rules in the knowledge base. Handed-over conversations appear in a queue a staff member can see, with the conversation and the customer's details. Once handed over, the assistant stops replying.
6. **Deployed.** Running somewhere we can reach during the walkthrough. If deployment fails, demo it locally and explain what went wrong.
7. **A README** covering how to run it, how to deploy it, your design decisions, known limitations, and what you would do next.

Extensions — choose at Checkpoint 1

Pick the extensions you will attempt, if any, and tell us why. Choosing well matters more than choosing many.

- **Agent view** — a human agent can read and reply in the handed-over conversation
 - **Payment allocation** — account answers reflect payments that have been received but are not yet in the loan book
 - **Combined answers** — questions that need both the customer's data and a policy, such as late penalties or early settlement amounts
 - **Staff mode** — Kopano managers can ask questions about the loan book in plain English
 - **Tests** — automated checks that the assistant behaves correctly
 - **Audit log** — a record of what the assistant said, to whom, and on what basis
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Checkpoints

Checkpoint 1 — 10 minutes

Tell us:

- The problem as you understand it, and anything in the materials that stood out to you
- What you will build: the core, plus the extensions you have chosen and why
- The biggest risks to getting it done in two hours
- What you asked the client, and what you learned

Checkpoint 2 — 10 minutes

Show us — a sketch on paper or a whiteboard is fine:

- The architecture: the components and how they connect
 - How you will load and query the data
 - How retrieval over the knowledge base will work
 - How verification and handover will work
 - Where and how you will deploy
 - What "done" looks like at the end of the two hours
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Walkthrough — 30 minutes

1. Demo your assistant
 2. We will chat with it as Kopano customers, using our own questions
 3. We will ask about your decisions and trade-offs
 4. We may change something and ask you to respond
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Rules

- Use any language, framework, database, vector store and hosting platform you like
- AI coding tools are allowed and expected. You must be able to explain every part of your work, including code an AI tool wrote
- Use the provided API key for any LLM and embedding calls
- Ask the client or the interviewers questions at any time
- You are not expected to finish everything. Tell us clearly what works, what doesn't, and why
- Commit or save your work regularly